

SMRITHI CHEMBATH

smrithichembath@gmail.com • (682) 336-4458 • Arlington, TX • [LinkedIn](#) • [GitHub](#)

SUMMARY

Data Scientist with 2+ years embedded inside e-commerce marketplace business units, leading the analytical function end-to-end, from framing the ambiguous business question to driving the decision with leadership. At a 10M+ user marketplace, built automated Python/SQL pipelines that cut insight delivery time by 75%, served as the go-to analytics partner for marketing and strategy teams, and was promoted from intern to analyst for taking on scope beyond my role. Comfortable with SQL, Python, first-principles problem framing, and communicating findings directly to senior stakeholders.

SKILLS

Languages & Tools: SQL (PostgreSQL, MySQL, Amazon Redshift, BigQuery), Python (Jupyter, Google Certified), Excel, Stata, GitHub

Analytical Approach: First-Principles Problem Framing, Hypothesis-Driven Analysis, A/B Testing & Experimentation Design, Regression & Time-Series Forecasting (SARIMA/SARIMAX), Cross-Validation & Model Benchmarking

Machine Learning & AI: Scikit-learn, XGBoost, LightGBM, BERT/NLP

Data & Visualization: Power BI (Microsoft Certified), Tableau, Pandas, NumPy, Databricks, Seaborn

Ownership & Communication: End-to-End Analytics Ownership, Executive/Leadership Communication, KPI Definition & Dashboarding, Stakeholder-Facing Recommendations

PROFESSIONAL EXPERIENCE

Research Scientist

April 2025 - Present

University of Texas at Arlington

- Led end-to-end analysis for a healthcare workforce hiring survey, defined the research questions, cleaned and structured large-scale datasets in Python and STATA, and identified staffing trends now informing policy reporting and academic publication.
- Built structured data pipelines that removed a recurring bottleneck in the research cycle, improving reproducibility without being asked.

Business Intelligence - Data Analyst

March 2023 - July 2024

Bayut & Dubizzle, Dubai, UAE

- Built automated Python and SQL workflows for Property and Motors divisions, cutting insight delivery from 3-4 weeks to 1 week (a 75% reduction) and becoming the go-to analytics partner for marketing and strategy leadership.
- Analyzed campaign and revenue data across a 10M+ user marketplace, applying regression and trend analysis to uncover pricing, growth, and efficiency opportunities; served as the primary analytics point of contact for marketing and strategy leadership.
- Defined and built the KPI dashboards leadership used to track Property and Motors listing performance, deciding which metrics mattered rather than just reporting what was asked for.

Business Intelligence - Data Analyst Intern

September 2022 - March 2023

Bayut & Dubizzle, Dubai, UAE

- Delivered 5+ dashboards across three business verticals end-to-end, independently managing scope as requirements evolved, work that led directly to a full-time promotion.
- Promoted from Intern to full-time Data Analyst after independently expanding scope beyond assigned tasks, surfacing revenue and operational trends across Property, Motors, and Classifieds verticals.

Machine Learning & Big Data Intern

August 2021 - January 2022

FlyDubai, Dubai, UAE

- Built decision tree models for operational disruption management, contributing to a ~50% reduction in flight reassignment time during cancellation scenarios.
- Incorporated cabin-class prioritization constraints (Business and First Class) into model logic to improve reassignment accuracy on large-scale flight operations data.
- Presented modeling insights directly to senior leadership, translating technical output into decisions they could act on immediately.

EDUCATION

Master of Science in Data Science

2024 - 2026

University of Texas at Arlington - GPA: 3.9/4.0

Bachelor of Engineering in Computer Science

2018 - 2022

Birla Institute of Technology and Science, Pilani - Dubai

RESEARCH & PROJECTS

Fraudulent Job Posting Detector | NLP, BERT, SHAP, Streamlit

- Built and deployed a fraud detection pipeline on 18K job postings (5% fraud rate), fine-tuning DistilBERT to 90% precision / 76% recall (PR-AUC 0.86) vs. a TF-IDF baseline; shipped SHAP-explained predictions via an interactive Streamlit demo.

Disaster Risk Forecasting | Power BI, XGBoost, SARIMA

- Engineered temporal and geospatial features from FEMA and weather datasets to train an XGBoost classifier achieving 85% accuracy; integrated SARIMA for trend forecasting and delivered findings via interactive Power BI dashboards.

Power Outage Forecasting | LightGBM, SARIMAX (INFORMS DMDA 2025)

- Built county-level outage forecasting models from NetCDF weather and historical data; implemented chronological cross-validation to prevent data leakage and benchmarked LightGBM vs. SARIMAX using MAE/RMSE and PR-ROC curves.